

Assignment 7: Who busts the Mythbusters?

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Exercise 1

- i. Look again at the three variables for this experiment in the `experimental_data` dataset. Which one is the explanatory variable and which one is the response variable? What value in the response variable should be classified as a success?

The explanatory variable is the group, the response variable is yawn, and a success is when a participant yawned, shown as a yes.

Exercise 2

- i. The number of yawns in the treatment group

10 yawns

- ii. The proportion of yawns in the treatment group

29%

- iii. The difference in number of yawns between the treatment and control groups

There are 6 more yawns in the treatment group

- iv. The difference in proportions of yawns between the treatment and control groups

$29 - 25 = 4$, there is 4% more in the treatment group.

Exercise 3

```
specify(yawn ~ group, success = "yes")
```

Exercise 4

```
hypothesize(null = "independence")
```

Exercise 5

```
generate(reps = 10000, type = "permute")
```

Exercise 6

```
calculate(stat = "diff in props", order = c("Treatment", "Control"))
```

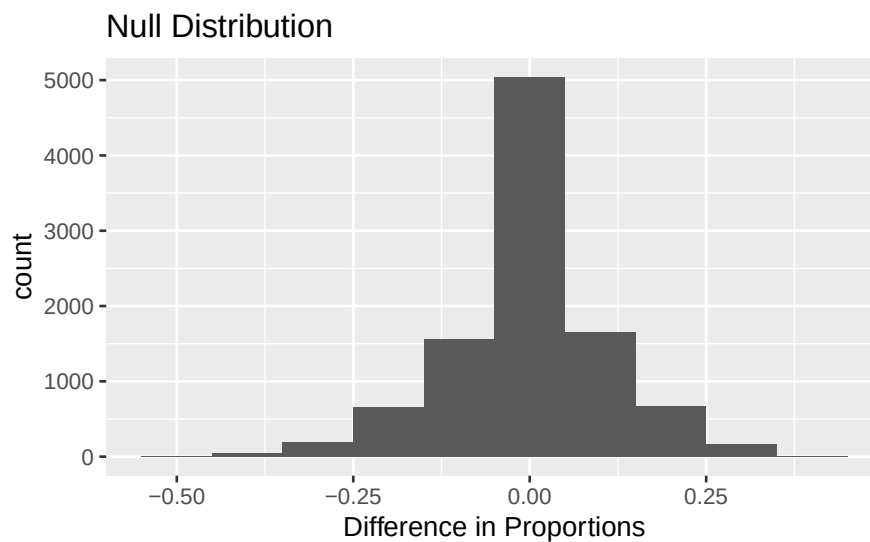
Exercise 7

```
yawn_null <- experimental_data %>%  
  specify(formula = yawn ~ group, success = "yes") %>%  
  hypothesize(null = "independence") %>%  
  generate(reps = 10000, type = "permute") %>%  
  calculate(stat = "diff in props", order = c("Treatment", "Control"))
```

Exercise 8

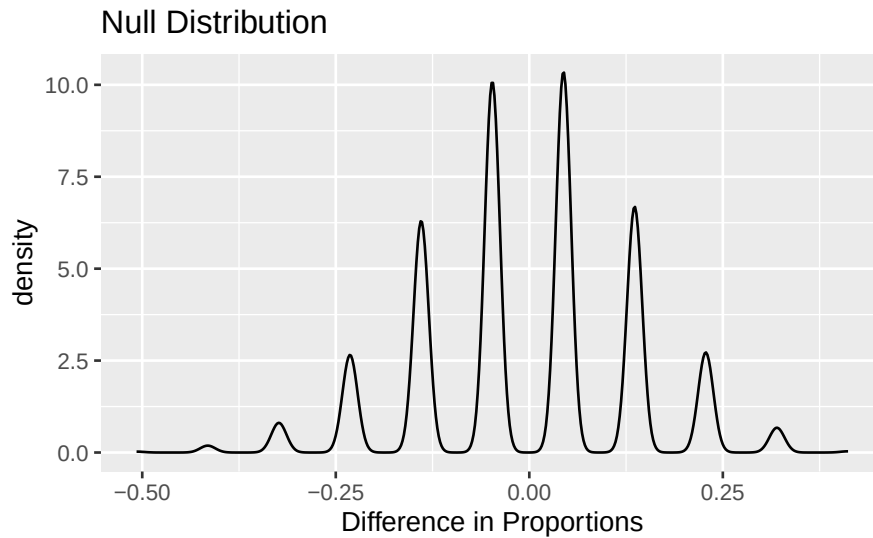
i.

```
ggplot(yawn_null)+  
  geom_histogram(aes(x=stat), binwidth=0.1)+  
  labs(title="Null Distribution", x="Difference in Proportions")
```



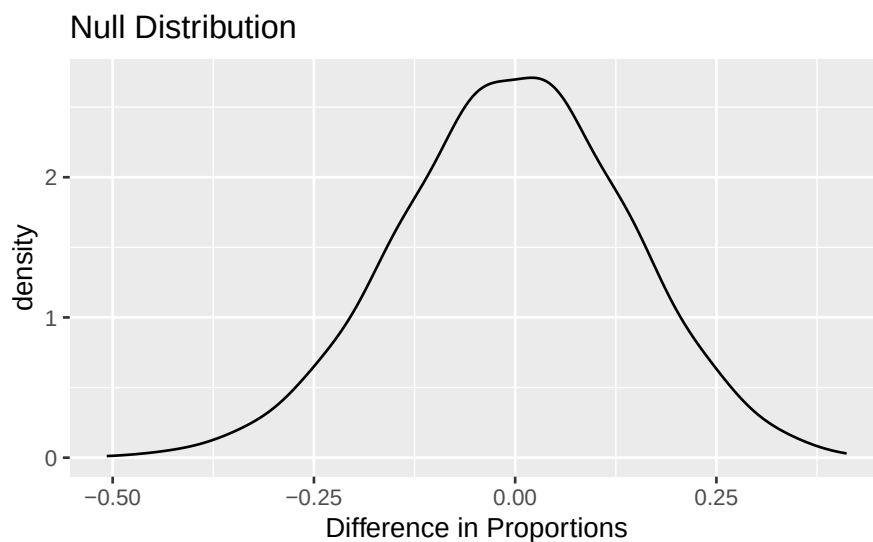
ii.

```
ggplot(yawn_null)+  
  geom_density(aes(x=stat))+  
  labs(title="Null Distribution", x="Difference in Proportions")
```



iii.

```
ggplot(yawn_null)+
  geom_density(aes(x=stat), adjust=5)+
  labs(title="Null Distribution", x="Difference in Proportions")
```

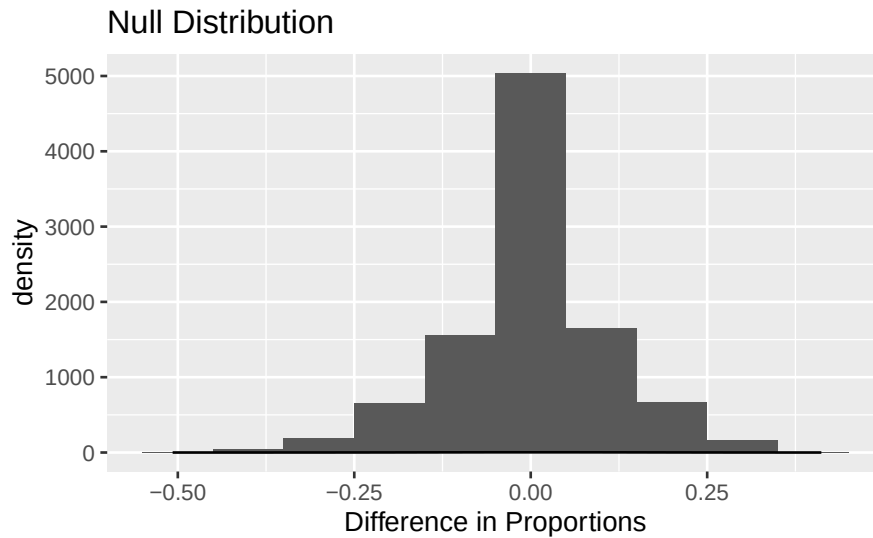


iv. Where is the center of each distribution? Does this make sense given what you know about the data (i.e. would you expect the difference of proportions to be centered on this value)?

The center is around 0, which makes sense since the null hypothesis assumes no difference in proportions.

Exercise 9

```
ggplot(yawn_null)+
  geom_histogram(aes(x=stat), binwidth=0.1)+
  geom_density(aes(x=stat), adjust=5)+
  labs(title="Null Distribution", x="Difference in Proportions")
```



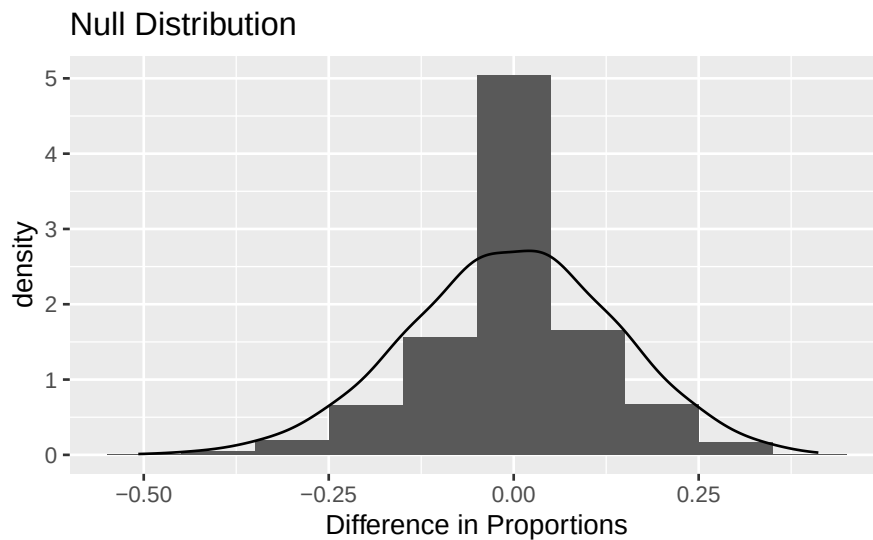
iv. Does the plot look okay, or do you notice a problem?

The density line is flat.

Exercise 10

```
ggplot(yawn_null)+
  geom_histogram(aes(x=stat, y=..density..), binwidth=0.1)+
  geom_density(aes(x=stat), adjust=5)+
  labs(title="Null Distribution", x="Difference in Proportions")
```

```
## Warning: The dot-dot notation (`..density..`) was deprecated in ggplot2 3.4.0.
## i Please use `after_stat(density)` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



Exercise 11

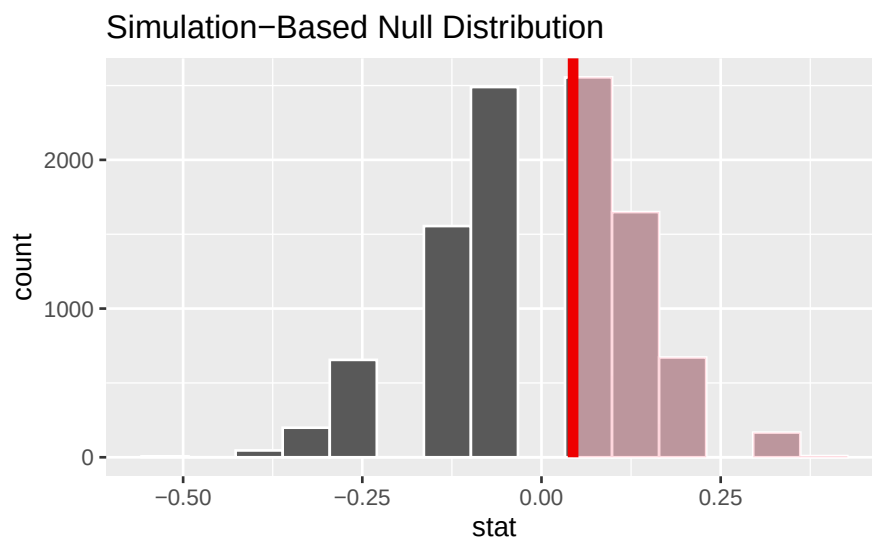
```
set.seed(42)

yawn_null <- experimental_data %>%
  specify(formula = yawn ~ group, success = "yes") %>%
  hypothesize(null = "independence") %>%
  generate(reps = 10000, type = "permute") %>%
  calculate(stat = "diff in props", order = c("Treatment", "Control"))

yawn_obs_stat <- experimental_data %>%
  specify(formula = yawn ~ group, success = "yes") %>%
  calculate(stat = "diff in props", order = c("Treatment", "Control"))

p_value <- yawn_null %>%
  get_p_value(obs_stat = yawn_obs_stat, direction = "right")

yawn_null %>%
  visualize() +
  shade_p_value(obs_stat = yawn_obs_stat, direction = "right")
```



p_value

p_value
0.5049

- Based on these results and a significance level of $\alpha = 0.05$, conclude whether or not we have sufficient evidence to reject the null hypothesis. Justify your answer.

The p-value is **0.5049**, which is greater than **0.05**, this means that there isn't sufficient evidence to reject the null hypothesis. The effect could be due to chance.

Academic Integrity statement

Chatgpt and the teachers guidance from SLACK was used to complete this assignment.

<https://chatgpt.com/share/6709df69-8d3c-800f-beac-838a4e803347>